

01

INJURY MANAGEMENT





PHOTO: MARTIN MOBRÄTEN

Shaken not Stirred aka Right Martini (V12), Hueco Tanks, USA.

HANDLING OF ACUTE SOFT TISSUE INJURIES AND OVERUSE INJURIES

We differentiate between acute injuries and overuse injuries because these different types of injuries require different approaches to treatment. However, there are some common factors for treatment and prevention, and in this chapter we will look at the management of acute injuries and overuse injuries from various perspectives.

We will examine the principles for managing acute injuries, and look at how physiological, psychological and social factors affect injuries; factors such as load management, strength training, recovery, sleep and nutrition. This chapter lays the foundation for chapter 2 which looks at specific climbing-related injuries (page [38](#)).

ACUTE SOFT TISSUE INJURIES

FEBRUARY 2019. *It is my last day in Hueco Tanks. I have never been to a better bouldering area. We have climbed a lot, and my body has taken a beating over the last ten days. Nevertheless, I feel good and ready to crush today's wishlist on East Mountain. After a quick – too quick – warm-up I go for a flash attempt on Mo Mojo (V11/8a), but fail miserably. I work through the sequence; the crux is pinching hard with my right hand while placing a high left heel hook by my left hand before deadpointing up to a left hand sidepull. The heel hook is high and close to my body, and I can feel that I really need to pull on it to get my body around the roof that the problem climbs along. However, it feels like something I've done hundreds of times before, so my surprise is great when on my next attempt I hear three distinct pops from my left hip and find myself sitting on the pad below.*

'What happened?' my friends ask.

I don't have an answer at the moment because, honestly, I don't feel much pain. Endorphins have that effect on us.

'Did you hear that?' I reply, as I check if my left hip is still where it should be.

It's the start of the last day, so it would take a lot for me to stop climbing today. And there isn't really any pain. I ignore the sneaking suspicion of a hamstring injury and move on to the next boulder. After a long day of climbing, a sandstorm forces us back to the car, and only then do I notice that I am limping. The next day, my left thigh is stiff all the way up to my buttock. The flight home from El Paso to Oslo is truly one for the books. Back home, what I initially suspected is confirmed. An injury in the hamstring muscles and tendon attachment at the sitting bone.

When tissues in our bodies are suddenly exposed to forces they cannot withstand, they become damaged to varying degrees. This tissue damage undergoes a healing process through three phases – *inflammation* phase, *repair* phase and *remodelling* phase – where the outcome of each phase affects both the next phase and the final outcome. This means that we actually want inflammation – an inflammatory response – even though, over the years, many hypotheses and models have been proposed which support the suppression of this response.

In this book we will focus on soft tissue injuries – injuries related to muscles, tendons and ligaments. Bone tissue injuries, such as acute fractures or stress fractures, are approached differently, primarily through stabilisation and immobilisation to ensure proper healing and stability.

In 1978, Dr Gabe Mirkin introduced RICE – Rest Ice Compression Elevation – as an acronym for managing acute injuries, with its goal being to reduce swelling while dampening inflammation, thereby accelerating the healing process. RICE quickly became indoctrinated in healthcare settings and society at large; it is still common practice for ice packs or compression bandages to be used immediately after someone sprains an ankle.

Like many theories, however, RICE has not stood up to scientific scrutiny over time. Indeed, rest along with cooling has been specifically shown to have negative consequences for the outcome of the inflammation phase. As W.B. Leadbetter put it: *'Inflammation can occur without healing, but healing cannot occur without inflammation.'*

WHAT DOES 'INFLAMMATION' MEAN?

The Latin origin of the word 'inflammation' is *inflammare*, which means 'set on fire'. It describes the symptoms of the inflammatory process: redness and swelling with heat and pain – *rubor et tumor cum calore et dolore*.

A common misconception is that the swelling following an acute soft tissue injury is synonymous with inflammation. However, the swelling represents the accumulation of waste products produced during the inflammatory process which have not yet been transported away from the site of the injury. This transportation is carried out by the lymphatic system, which is a one-way system where fluid is pumped through lymph vessels back to the heart. This pumping function relies on muscle activity, so actively using muscles is the best way to reduce swelling. Immobilising an injured body part therefore does not contribute to reducing swelling; this inactivity also leads to muscle deconditioning, resulting in a poorer final outcome. And it is not only muscles that suffer due to inactivity. Tendons, ligaments and bones also weaken if they are not regularly stressed, so being completely immobile after an acute soft tissue injury is not something we should do.

Nowadays, we understand that we actually want and need an inflammatory process following an acute soft tissue injury, yet at the same time we want to reduce swelling and maintain muscle activity and load. We want to avoid situations that can worsen the injury and we want to avoid the use of anti-inflammatory medications, aka NSAIDs (non-steroidal anti-inflammatory drugs). This development of our understanding over time has led to an evolution from RICE to PRICE to POLICE to MEAT to MOVE*, finally resulting in the latest acronym, **PEACE & LOVE**.

The first part, **PEACE**, applies during the acute phase in the first few days following an injury.



*RICE = Rest Ice Compression Elevation

PRICE = Protection Rest Ice Compression Elevation

POLICE = Protection Optimal Load Ice Compression Elevation

MEAT = Movement Exercise Analgesics Treatment

MOVE = Movement Options Vary Ease

PROTECT

This involves avoiding worsening the injury. If we injure a finger while bouldering, twist an ankle or experience a sudden jerk in a shoulder, it makes sense to end the session early. Although we don't want complete rest for the injured body part, we can limit its use, the load it is under and its range of motion to reduce the risk of worsening the injury. I see that I failed miserably at this first step when I injured my leg in Hueco, although I did switch boulders after the injury.

ELEVATE

Keeping the injured body part above heart level means that fluid accumulated during the inflammation phase can more easily flow back towards the heart via lymphatic system drainage, thus reducing swelling.

AVOID ANTI-INFLAMMATORY MODALITIES

As described on the previous page, we are aiming for a successful inflammatory phase, rather than trying to suppress it. Therefore, we should avoid NSAIDs and cooling measures. Such interventions may reduce pain, which can certainly be desirable in some cases, but they will have no positive effect on the inflammation phase or the final outcome.

COMPRESS

Despite a lack of documentation on the effectiveness of compression bandages, they may help to reduce swelling by providing support around an injured body part. They also appear to be beneficial for quality of life during the period following, for example, an ankle sprain.

EDUcate

In a world full of internet and YouTube gurus, infinite Google search results, quick fixes and high-tech solutions, it can be difficult to know who to listen to and what information to rely on. Providing good answers related to the type of injury, timescales, prognosis and what a patient can do during the healing process is an important part of a healthcare professional's job, and this is the information we should expect when seeking professional help. The late English physiotherapist Louis Gifford suggested that there are four questions all patients want answered when seeking help:

1. What is wrong with me?
2. How long will it take to get better?
3. What can I (the patient) do?
4. What can you (the healthcare professional) do?

Good answers to these questions, based on the best available knowledge, will give us a quicker start to an active approach to healing and increase our confidence in the process. It will also reduce the need for treatment modalities and medication, while allowing the rehabilitation to progress naturally in the best possible way. Being able to answer all these questions was crucial when it came to handling my own hamstring injury.

While **PEACE** is reserved for the first few days following injury, soft tissue injuries then require **LOVE** in the period following the acute phase.

LOAD

All tissues in the body repair themselves and build up capacity through loading. The underlying mechanism for this is called 'mechanotransduction', and it is now well known how mechanical stimuli increase muscle, tendon and ligament load-bearing capacity. Optimal loading during this phase involves loading the tissue without worsening the symptoms, such as pain or swelling, afterwards.

OPTIMISM

Injuries rarely come at convenient times, and they can negatively affect quality of life and mood. Mental factors such as fear of movement, depression and anxiety can be major obstacles in the rehabilitation process, with a direct negative impact on the final outcome. There isn't necessarily a close relationship between injury severity and symptoms experienced, but there is a much stronger connection between thoughts and emotions and symptoms. It's about how we feel, but also how we handle those feelings. An optimistic mindset leads to better outcomes and prognosis, so we need to focus on the positive in a tough situation. This is one of the reasons why this book has a dedicated section on movement optimism (pages [147–148](#)).

VASCULARISATION

Good fitness isn't necessarily a performance factor for climbers, but after an injury it is important to start general strength and conditioning training early. This increases blood flow throughout the body, potentially reducing pain and improving motivation to continue training. Such forms of exercise also make us feel actively involved in the recovery process; with a finger injury, there is actually a lot of other training we can be doing. Not only does this positively impact the rehabilitation process, but it also makes us stronger and improves our overall physical condition.

EXERCISE

Exercises that target injured tissue structures are the cornerstone of the rehabilitation process. As already explained, exercises provide the specific load to the tissue which optimises healing. Exercises that contribute to regaining strength, full range of motion and confidence in the injured body part are crucial, and should be started as early as symptoms allow.

The most important message you should take away from this section is that **acute soft tissue injuries do not require prolonged periods of rest**. They involve an acute inflammatory phase which is required for optimal healing, but it is then absolutely necessary to load the tissue from an early stage to ensure the best possible outcome. The most common climbing-related injury is an acute pulley rupture in a finger, and the principle of **PEACE & LOVE** applies greatly to this type of injury. Keep this in mind when you read the section on pulley injuries on pages [41–51](#).

OVERUSE INJURIES

Whereas acute injuries have a clearly defined time of injury (often a specific event), both the cause and the time of onset are less clear in the case of overuse injuries and conditions. Simply put, overuse injuries can be the result of exercising:

TOO MUCH, TOO OFTEN, TOO SOON and with TOO LITTLE REST.

However, it is difficult to be certain about what is too much or too often. Therefore, we consider the load from various perspectives – physiological as well as psychological and social. Let's start with the physiological aspect.

All tissues in the body adapt in response to loading. This is why muscles become larger and can generate more force, why tendons and ligaments become thicker and stiffer, and why bones build more bone tissue. Overall, loading allows us to develop strength and resilience against higher forces. To stimulate this adaptation process, we must expose our bodies to stimuli which they are not already capable of handling. This applies to all types of training – from purely physical, to technical and mental training. If we exceed what the tissues can withstand without then providing sufficient rest so they can adapt, no adaptation will occur; instead, this repeated tissue breakdown may eventually lead to symptomatic tissue damage.

We can use our fingers as an example. By climbing on steeper wall angles and using smaller holds, we can expose our muscles, tendons, ligaments and bones to forces that can make us stronger. This leads to thicker and stiffer pulleys and tendons, thicker bones and broader joint surfaces if we rest adequately between sessions. However, without sufficient recovery, we may experience adverse reactions such as elbow tendinopathy, tenosynovitis within the flexor tendon sheaths and synovitis in the finger joints. In younger climbers who are still growing, excessive loading can cause changes and potentially fractures in the growth plates of their finger bones (read about my own experience of this on page [8](#)). It is difficult to determine exactly where the limit is, but this is the key. If we can control the load, so that there is a balance between stress and recovery, we can both reduce the risk of injury and manage it better if and when it does occur. Rehabilitating an overuse injury is impossible without adjusting the load that caused it; therefore, all rehabilitation programmes should have load management as a fundamental principle.

A good place to start is the **LOVE** protocol (see [previous page](#)). Load management is not about a complete unloading, but rather about adjusting the different elements that make up the load. By adjusting both volume (how much we train) and intensity (how hard we train), we can continue training and climbing if we also ensure sufficient rest. Paradoxically enough, loading is a crucial part of managing overuse injuries. In most cases, there are many things we can still train that will have positive effects on our physical and mental well-being. A painful finger may lead us to reduce specific finger training and avoid steep crimping for a short period, but we can still practise technique on larger holds with gentler wall angles, practise clipping techniques while lead climbing, and work on arm strength and basic conditioning.

This way, we provide some reduction in load for the affected finger while simultaneously improving other aspects of our climbing through general training. This general approach to training also has a positive effect on reducing recovery time for an injured finger, and generally has a positive psychological response because we are actively doing something towards getting better.

'IF YOU GO TOO HARD ON YOUR EASY DAYS ... SOON YOU WILL BE GOING TOO EASY ON YOUR HARD DAYS.'

KEIJO HÄKKINEN, WORLD-LEADING RESEARCHER IN POWER TRAINING



A good method for becoming aware of and monitoring training load over a period of time, such as a week, is labelling sessions as red, yellow or green. Red represents the hardest sessions, green represents the easier sessions, and yellow sessions fall in between. A hard bouldering session for two hours would be considered a red session, and looking back over a week it is possible to see how many red sessions there have been. If there are lots of these sessions within a week, this is not a sustainable training strategy. To improve our climbing ability, we must climb a lot, which means that we must vary our training so it is more sustainable. This could still mean bouldering for two hours, but instead on gentler wall angles, or focusing on technical balance problems. It may not be as physically demanding, but it requires more technical skill.

By combining more general physical training with specific training for an affected body part, we can avoid large fluctuations in the loading rhythm. If a climber is used to climbing and training three times a week but then takes three weeks off completely, there will be a drop in training load. When they start again, they should build up towards their previous level of training load; however, this pattern of peak-drop-peak can contribute to recurring issues. We should strive to maintain a relatively steady workload with only short and moderate fluctuations periodically to vary the load. However, illness, injury, holidays and life in general can cause a drop in training intensity that deconditions us physically. Few of us are diligent enough to gradually rebuild our condition following these periods, and so this often leads to us doing too much, too soon.

Being aware of this can serve as an injury prevention measure: if we incorporate brief sessions of finger and upper-body strength training during times when we have less time or opportunity to climb, we will be much better prepared when we resume climbing again.

INJURY PREVENTION: MORE THAN JUST TRAINING?

Unfortunately, when we train to become better climbers, injuries and pain are almost inevitable parts of the process. This is why it has been suggested that the term ‘injury prevention’ is replaced with the term ‘risk reduction’ – because it is impossible to prevent injuries one hundred per cent. A preventive measure can at best reduce the risk of an injury occurring, but even if we do everything right, there is still an element of chance involved. Bad luck. Shit just happens. Regardless of which term we choose to use, there are steps we can take to reduce our risk of injury, and, if we manage to do the fundamental things well, we have at least done what we can to try to stay injury free.

First and foremost, it is important to distinguish between *external* and *internal* risk factors. External factors are things like whether the safety equipment is in order, whether there are holes in the crash pads or if they are placed next to each other properly so there is no gap which could roll an ankle, whether the last move on a boulder is a sideways dyno, or if your training wall only has small crimps on it. Internal factors concern us as people and how we respond to stressors, and these factors are primarily what I want to highlight in the rest of this chapter.



TRAINING LOAD: A PART OF THE WHOLE

Training load is one part of the *total load* we are exposed to, and how much training is too much is very much determined by the individual and depends on several factors. A training process consists of *external load* – volume, intensity, exercise selection and type of training – and *internal load* – how we physiologically respond to the external load and what the outcome is.

This is a simplification of training load, and when we train both hard and frequently, the external training load is generally high. How we respond to this load is influenced by various factors, such as our individual characteristics, training status and age, fitness level, overall physical and mental health status, nutritional status, genetic material and our environment. Therefore, the same external training load can have different outcomes for different people, making it difficult to predict whether a training plan will have the desired effect or if the training load may become excessive. Some parameters are also easier to control than others. For example, it's easier to measure and control deadhangs – hanging motionless from straight arms on a fingerboard – by looking at weight, edge depth and hang time, compared with assessing the load during a climbing session. However, when climbing we can still consider the number of moves per session as an indicator of volume, while wall angle and hold size can be used as indicators of intensity during each session. Many moves on steep walls with small holds will result in a high overall external training load. By reducing volume during such sessions, we can achieve quality workouts at a high intensity, while performing many moves on less steep wall angles with larger holds would result in a session that is higher in volume but lower in intensity. The obvious challenge with

climbing is that it's so enjoyable to climb both hard and a lot, that we often end up doing multiple consecutive sessions with consistently high levels of external load.

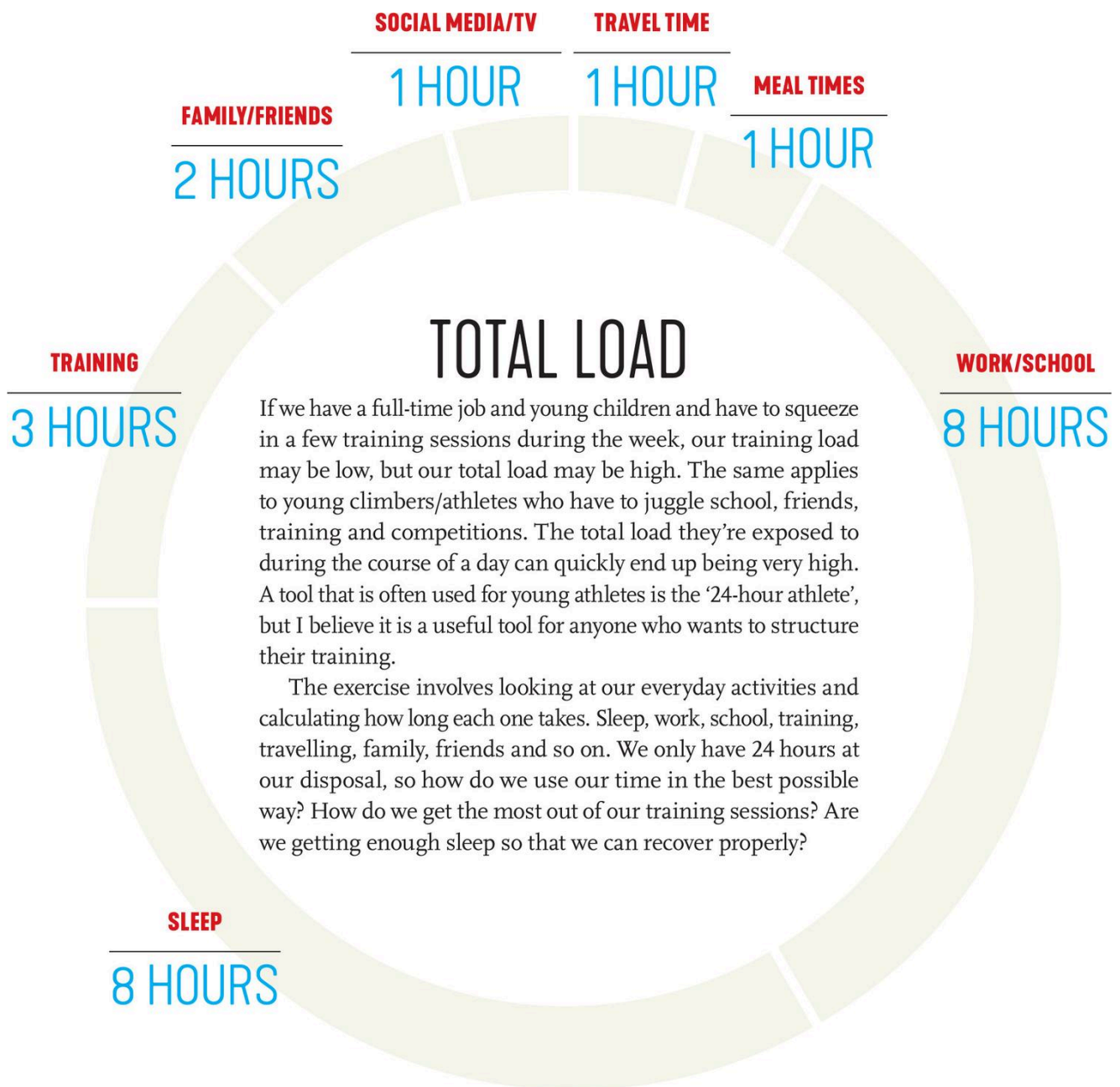
Several models have been proposed to predict injury risk based on changes in load, the most popular being the Acute: Chronic Workload Ratio (ACWR). This model calculates the training load for the last week (acute training load) and divides it by the average training load for the previous three to six weeks (chronic training load). The greater the difference between these two loads, specifically if the acute training load is higher than the chronic, then the higher the ratio and thus the risk of injury. Although this model quickly gained popularity due to its simplicity, from a scientific perspective it has significant flaws that prevent it from being used as anything more than a measuring tool for external load. There are so many factors other than just the relationship between acute and chronic training loads that contribute to injury risk, that all models are ultimately too simplistic. However, we should still consider the basic principle of managing training load as a risk-reducing measure.



When discussing overuse injuries and load management, we must also consider the concept of *total load*. Unlike training load – which refers to how much and how hard we have trained – total load encompasses *everything* we are exposed to in life, including work, childcare, illness, moving house and so on. There will be periods with intense training, periods with poor sleep quality and periods with increased emotional stress. For example, most parents of young children would agree that this phase in life can be demanding, making it difficult to tolerate the same level of training load as before if they're only sleeping in two-hour intervals while simultaneously working and taking care of a small child or children. Changes in life situations, such as periods of high work pressure or negative relationships, will also increase total load, thereby reducing our capacity to handle additional physical strain. Being aware of how all these elements are interconnected and influence one another gives us an opportunity to adjust our training loads during certain periods, while accepting that it's not always possible to achieve our desired volume.



Having children is wonderful, but most parents agree that it can also be exhausting. Periods of reduced sleep affect both training quality and recovery. Considering the wants and needs of another human being affects how we prioritise our own time and makes us think twice about training time and volume. If we're lucky enough to have children who want to join us on a trip, we can also look at it as a long-awaited opportunity to take a slightly longer rest between attempts, so we might be able to complete our project after all! Stian and Kasper (then aged six) enjoying lunch on the pad by Bruno Bloc in Magic Wood, Switzerland.



SLEEP AND NUTRITION

Most of us have experienced the effect of poor sleep quality on our bodies, and so it should come as no surprise that insufficient sleep can be a risk factor for sports injuries. Insufficient sleep leads to an increase in physiological demand during physical and cognitive activity, which results in more rapid fatigue compared to if we had slept better and for longer.

Adolescents between the ages of 12 and 18 who sleep for less than eight hours a night are reported to have a significantly higher injury risk compared with their peers who sleep for eight hours or more. Considering that the amount of sleep generally decreases during adolescence, and the risk of sports injuries increases during the same period, we can make an assumption that reduced sleep over time is a risk factor.

Reduced sleep also directly affects how well and how quickly we're able to recover following a training session, thus having consequences for how well we tolerate training load over time. The external load may be the same, but the internal load is significantly higher and will eventually lead to reduced performance and potentially overuse injuries. Additionally, considering that reduced sleep is associated with poorer nutritional status, we now have two important factors negatively affecting our recovery time.

The Norwegian Olympic and Paralympic Committee and Confederation of Sports has developed nutritional guidelines for injury periods (see [next page](#)), but we can also use these guidelines to ensure a nutritional status that can reduce the risk of injuries associated with inadequate nutrition. The most important thing is to have

enough energy and carbohydrates available to meet the strain we put on our bodies, and this is particularly important for children and adolescents. A deficit in energy balance is not a sustainable strategy for either adults or youths who want to train extensively, and malnutrition is linked to both physical injuries and mental health problems. A diet consisting of regular meals that include fats, carbohydrates and proteins, as well as the recommended daily doses of fruit and vegetables, is a more than sufficient nutrition strategy for most people to help reduce their risk of injuries.

So, to all of you young and promising athletes – and to all you hard-working parents – **get enough sleep and eat well!**

RECOVERY NUTRITION

EAT ENOUGH FOOD – DON'T LOSE WEIGHT

This is not the time for dieting!

- Important for maintaining muscle mass.
- Protein intake of approximately 1.6–2.5 grams per kilogram of body weight per day.
- Increased energy requirements during and after surgery.



EAT REGULAR MEALS

Breakfast and evening meals are important meals!

Aim for 20–40 grams of protein per meal.

- Regular protein replenishment will stimulate protein synthesis and reduce muscle loss.
- Evening meals should be high in protein.



AVOID NUTRITIONAL DEFICIENCIES



- Check vitamin D and iron status.
- Include antioxidant-rich foods in your diet, such as fruit, berries, vegetables, nuts, etc.
- Increased calcium requirements in the event of fractures and missing periods.



OMEGA 3

2–4 grams per day (two to four capsules per day)

- Increases muscle-building sensitivity in muscles.
- Reduces inflammatory response.



ENOUGH SLEEP, LITTLE ALCOHOL

- Good recovery routines are very important.



DIFFERENCES IN AGE AND GENDER

‘Everything hurts a little more now. My finger joints ache, my back is stiff and I feel more beaten up after a good climbing session. In my younger years, I could do a one-arm pull-up straight out of bed, but now it feels like my arms are going to snap when I do a pull-up without warming up.’

It is not uncommon for me to hear statements like this at my clinic, or to overhear them at the climbing wall or crag. As we transition from childhood and adolescence into adulthood, through middle age and beyond, there will be differences in how our bodies adapt to different forms of exercise, and recovery times after strength training and high-intensity interval training will vary throughout our lives. There are also differences between genders, further highlighting the fact that we cannot simply use the same training or rehabilitation plan for every individual. When I compare myself as a 40-year-old to myself as a 20-year-old, there is a significant difference in the training volume I can now handle and in how long it takes me to recover; in practice, we see that many people have similar experiences.

However, it is not possible to be definitive about how age alone affects recovery time. The level at which we climb and how long we have been training will influence both the effectiveness of a training plan and our recovery time after a workout. Therefore, it is not necessarily true that a well-trained middle-aged climber recovers significantly more slowly than a 20-year-old beginner.

Knowing oneself and one’s own body is crucial for being able to continuously regulate training load through different periods in life.

MENSTRUATION AND INJURIES

Research on this topic is so scarce, that Martínez-Fortuny et al., the authors of the latest major review article in this field, commented: *‘The lack of scientific evidence on this topic is remarkable, and further research is needed in order to validate these hypotheses.’* Keeping this in mind, we can still say that variations in hormone production throughout the menstrual cycle are related to increased risk of injury, especially during ovulation.

Factors such as stiffness in connective tissue, neuromuscular control and strength also vary throughout the cycle, and changes in these factors can affect the risk of injury. It is obviously impossible at present to provide specific and general advice on training and injury risk throughout the menstrual cycle, but where an individual is in their cycle can be one of many factors influencing their injury risk.

Athletes are encouraged to monitor their cycle phase and any symptoms, so that they can better adjust their workload during different phases. Having a better understanding of one’s own cycle will contribute to making better medical decisions, and being aware of this allows for the adjustment of the intensity and training methods according to what one knows is right for one’s body.

The increased openness about training and menstruation is gratifying, because only through more knowledge, improved understanding and better communication among athletes, coaches and healthcare professionals can we optimise health and performance for female athletes.

WARMING UP

In sports that include a structured warm-up programme which targets the specific demands of the sport, it is generally observed that such programmes help reduce the incidence of injuries. This has been demonstrated in sports like football and rugby, and there is reason to believe that it should also apply to climbing. The physiological processes brought about by warming up make muscles, tendons and connective tissues more elastic – thus more resistant to strain, and increase muscle activation, improve coordination and mentally prepare us for activity. For example, it has been shown that 100–130 moves on a climbing wall makes the pulleys more elastic. By performing some portion of this volume on a fingerboard, one can progressively and in control vary grip positions while increasing both load and rate of force development during warm-up, while simultaneously warming up the rest of the body as well.

The goal of our climbing or training session affects the final part of our warm-up. If we are going to have a finger-intensive and hard bouldering session, the last part of our warm-up should include fast pull-ups, jumps to edges and some sets of heavy deadhangs with a 3–5-second hang time. If the focus of the session is more technical, working on elements like balance and weight transfer, the last part of the warm-up should be directed more towards coordination and balance to stimulate body awareness and improve motor-learning outcomes.

SUGGESTED WARM-UP EXERCISES

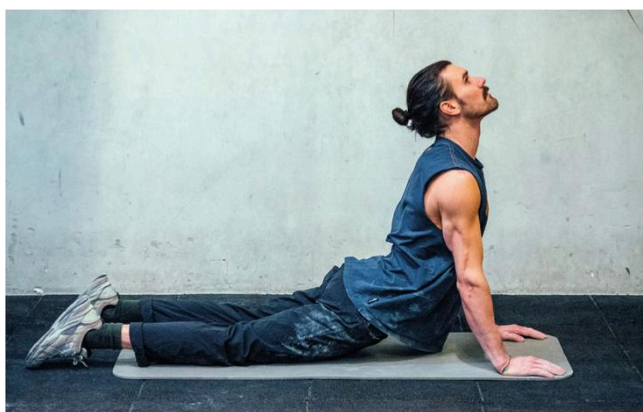
The aim of the warm-up programme is to get the whole body moving. The exercises presented here can serve as a starting point that you can adapt to your own needs, but I recommend including exercises that involve the upper body and shoulders, hips, legs and of course the arms and fingers. The exercises can be done as a circuit, with the first round easier than round two, so that you gradually increase the load and finish ready to climb.

FLOOR EXERCISES

10–12 REPETITIONS X2 SETS



CHEST BRIDGE Start sitting on your knees with your toes on the ground and your hands on your heels. Press your hips forwards, rotate your shoulders backwards and create a bridge with your upper body.



COBRA TO DOWNWARD-FACING DOG. Start lying on your stomach and lift your upper body by straightening your arms. Then push yourself backwards and bring your head and upper body in between straight arms.



COSSACK SQUAT. Start standing with a wide leg stance. Sit down over one foot while straightening the other leg. Stand up again and do the same on the opposite side.



SINGLE-LEG BRIDGE. Start lying on your back with one leg on a box, crash pad or similar. Press your heel against the surface and lift your body off the ground. Vary the angle of your knee and how much you rotate your leg out to the side.

HANGING EXERCISES

5 SECONDS' HANG: 5 SECONDS' REST X5 REPETITIONS X2 SETS



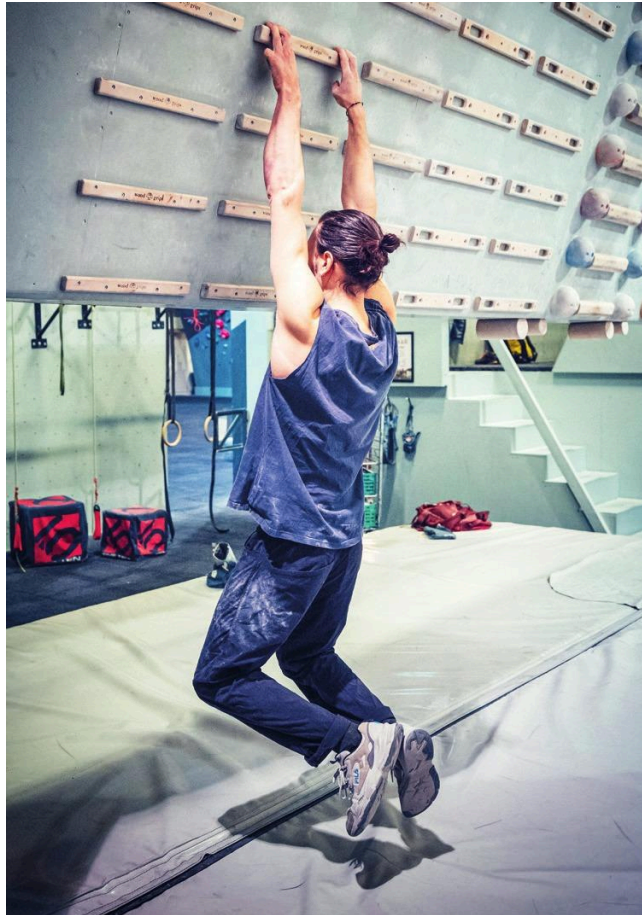
SCAPULAR PULL-UP. Hang passively from good holds. Pull yourself up by pulling your shoulders down and back, keeping your arms straight. Hold the top position for 5 seconds, slowly lower back down, rest for 5 seconds and repeat 5 times.



ONE-ARM ACTIVE HANG. Hang from one arm in the top position of a scapular pull-up and hold the position for 5 seconds; repeat 5 times with 5 seconds' rest in between reps.



DEADHANG, HALF-CRIMP GRIP. Hang from an edge with a half-crimp grip in the top position of a scapular pull-up for 5 seconds. Rest for 5 seconds, repeat 5 times. Alternatively, if it is too difficult to hang with your full body weight, you can stand on the ground and only take part of your body weight.



DEADHANG, THREE-FINGER OPEN-HAND GRIP. Hang from an edge with a three-finger open-hand grip in the top position of a scapular pull-up for 5 seconds. Rest for 5 seconds, repeat 5 times. Alternatively, if it is too difficult to hang with your full body weight, you can stand on the ground and only take part of your body weight.

STRENGTH TRAINING

It seems logical that if we become stronger in the areas where the demands are the highest, we will be able to withstand greater forces, thus reducing our risk of injury. From this perspective, all the strength training we do to become stronger climbers, who perform better, can also reduce our risk of injury.

However, here we face a dilemma: if we climb a lot, but at the same time extensively train finger, arm and upper-body strength, we will likely end up with such a high training load that our risk of injury will have increased. But this is a *dosage* problem, not a problem with the training methods themselves. While it's not surprising that self-reported injuries are associated with training facilities such as campus boards, fingerboards and system walls, these are just training installations – they tell us nothing about which training method has been used or at what dosage. An individually tailored strength training programme, aimed at improving finger, arm and upper-body strength while also adapting to the load from other climbing activities, should be a cornerstone in any training regimen – both for performance enhancement, but also as a risk-reduction strategy. By adapting the range of exercises, volume, intensity and frequency, strength training is also a fundamental component in rehabilitation, with the goal of enabling tissues to withstand movement and the stress to which they are subjected.

Therefore, strength training is central to both injury prevention and rehabilitation, where individualisation involves exercise selection and training dose.

Naturally, there is no universal programme, but Glasgow et al. advocate that an optimal loading regime should include the entire neuromuscular system. To accomplish this, strength training should:

1. target the appropriate tissues
2. ensure loading through functional joint range of motion
3. balance compressive, tensile and shear loading
4. vary the magnitude, direction, duration and intensity

5. incorporate neural overload
6. adapt to individual characteristics
7. be functional

If – based on these principles – we regularly train strength over time, research on a wide array of sports has shown that our injury risk can be reduced by up to a third. While climbing-specific research is currently lacking, it is not unreasonable to conclude that climbers can reduce their risk of injury through structured strength training.

PREVIOUS INJURY

The most significant risk factor for sustaining an injury is a previous injury. Some types of injuries, such as strains, tend to flare up again if we don't do a proper job of rehabilitating them. Other types of injuries can predispose us to new injuries in other parts of the body because we modify our technique and movement patterns to compensate, or because pain hampers force development and coordination. A rehabilitation process is therefore not complete until we have worked our way back to – and preferably beyond – the capacity we had before the injury. This also emphasises the need to start loading an injured body part early, and the necessity of taking sufficient time to return to climbing at our desired level. In this regard, the natural variation pattern that climbing provides us with is both a blessing and a curse. A blessing because there is always plenty of climbing we can do, but a curse because there is so much that we can climb, we can climb around the problem instead of addressing it properly. Fortunately, we can use this to our advantage by using climbing as both training and rehabilitation, as long as we are aware of exactly what it is that we are training for.

SUMMARY

Preventing all sports injuries is not possible, but we can influence various factors that collectively can reduce our risk of injury. In climbing, there are many external factors involved, such as checking that all safety equipment is in order, ensuring crash pads are positioned correctly and assessing fall risks. However, there are also many internal factors. I have chosen to highlight the fundamental internal factors associated with increased risk of injury in order to emphasise the message of doing the simple things well and to demonstrate how these factors are closely interconnected. For example, there will be times when we're sleep deprived, times when it's hard to eat both well and enough, and times when it's difficult to train as we'd wished or planned. This is completely okay, but we must make adjustments in our training load during these periods, so that the total load does not become excessive.